

Deep Learning for Environmental Monitoring: A Comprehensive Analysis of Object Detection and Semantic Segmentation Approaches for Litter Detection in the TACO Dataset

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Abstract. This report presents a comprehensive analysis of deep learning approaches for automated litter detection using the TACO (Trash Annotations in Context) dataset. We systematically compare object detection methods using YOLOv8 variants against semantic segmentation approaches using enhanced UNet architectures with attention mechanisms. Our experimental design encompasses three distinct YOLOv8 configurations and one advanced UNet implementation, focusing particularly on the challenging task of detecting small objects such as cigarettes. The report reveals that input resolution is a critical factor for small object detection, with 1280px resolution enabling a significant improvement in cigarette detection performance. Object detection significantly outperformed semantic segmentation (25.27% vs 9.43% mAP/mIoU), with YOLOv8m at high resolution achieving production-viable results.

Keywords: Object Detection · Semantic Segmentation · Environmental Monitoring · Small Object Detection · YOLOv8 · UNet · Computer Vision

1 Introduction

Environmental monitoring through automated computer vision systems has emerged as a critical application area for addressing global waste management challenges. The proliferation of litter in urban and natural environments necessitates scalable, automated detection systems that can assist in monitoring and cleanup efforts. This study focuses on the TACO (Trash Annotations in Context) dataset [1], which represents one of the most comprehensive collections of annotated litter imagery for environmental monitoring applications.

The primary challenge in automated litter detection lies in the extreme diversity of object sizes, from microscopic cigarette butts (2-10 pixels) to large bottles (200+ pixels), combined with severe class imbalance where background pixels constitute approximately 95% of the dataset. Traditional computer vision approaches struggle with such extreme scale variations and class distributions, making this an ideal testbed for modern deep learning architectures.

Our research addresses three fundamental questions: (1) How do object detection and semantic segmentation approaches compare for litter detection tasks? (2) What impact does input resolution have on small object detection performance? (3) Which architectural and training strategies are most effective for handling extreme class imbalance in environmental monitoring scenarios? This report contributes a systematic comparison of YOLOv8 variants and enhanced UNet architectures, with particular emphasis on optimization strategies for small object detection.

2 Related Work

Computer vision applications for environmental monitoring have gained significant attention in recent years. Object detection frameworks, particularly YOLO architectures [2], have demonstrated strong performance across various scales but traditionally struggle with very small objects. Semantic segmentation approaches using UNet architectures [3] have shown promise for pixel-level classification tasks but face challenges with extreme class imbalance.

Recent advances in small object detection have focused on multi-scale feature fusion [4] and high-resolution input processing. However, the computational trade-offs and optimal resolution selection for practical deployment remain underexplored, particularly in environmental monitoring contexts.

3 Dataset Analysis

3.1 Dataset Overview

The TACO dataset comprises 1,500 high-resolution images with 4,784 annotations across 59 distinct litter categories organized into 28 supercategories. For this study, we focused on the five most prevalent classes based on frequency distribution, representing the most common litter types in urban environments.

3.2 Class Distribution and Characteristics

Our analysis identified critical characteristics that directly impact model performance:

The dataset presents several critical challenges: (1) extreme class imbalance with background pixels dominating ($\sim 95\%$), (2) scale variation spanning two orders of magnitude, (3) environmental variability across lighting conditions and contexts, and (4) annotation quality variations affecting segmentation ground truth.

3.3 Dataset Distribution Analysis

Figure 1 illustrates the supercategory distribution in the TACO dataset, highlighting the extreme class imbalance that fundamentally challenges both detection and segmentation approaches.

Distribution Analysis Insights:

Table 1. Top 5 TACO Dataset Classes: Distribution and Characteristics

Class	Annotations	Percentage	Key Challenges
Cigarette	667	13.94%	Extremely small (2-10px), background similarity
Unlabeled litter	517	10.81%	Heterogeneous appearance, variable size
Plastic film	451	9.43%	Semi-transparent, deformable shapes
Clear plastic bottle	285	5.96%	Large but transparent, distinct edges
Other plastic	273	5.71%	Diverse subcategory, mixed scales

- **Cigarette Dominance:** Despite being the smallest objects (2-10 pixels), cigarettes constitute the largest class with 667 annotations
- **Scale-Frequency Paradox:** The most challenging class (cigarettes) is also the most frequent, amplifying detection difficulties
- **Heterogeneous Categories:** "Unlabeled litter" and "Other plastic" represent diverse subcategories with mixed visual characteristics
- **Clear Hierarchy:** Exponential drop-off in frequency creates training bias toward dominant classes
- **Background Overwhelming:** Combined target classes represent <50% of total annotations, with background dominating pixel-level analysis

3.4 Data Preprocessing and Split Strategy

We employed a stratified random split maintaining class distribution: 80% training (960+ images), 10% validation (120+ images), and 10% testing (120+ images). All images were filtered to contain at least one instance of the target classes, resulting in 1,200+ usable images.

4 Methodology

4.1 Object Detection: YOLOv8 Variants

We implemented three distinct YOLOv8 configurations to systematically analyze the impact of model capacity, input resolution, and training strategies:

Experiment 1: YOLOv8m Baseline

- Model: YOLOv8m (Medium variant)
- Input Resolution: 640×640 pixels
- Training: 40 epochs, batch size 16, AdamW optimizer
- Augmentation: Standard YOLO augmentation suite

Experiment 2: YOLOv8m High-Resolution

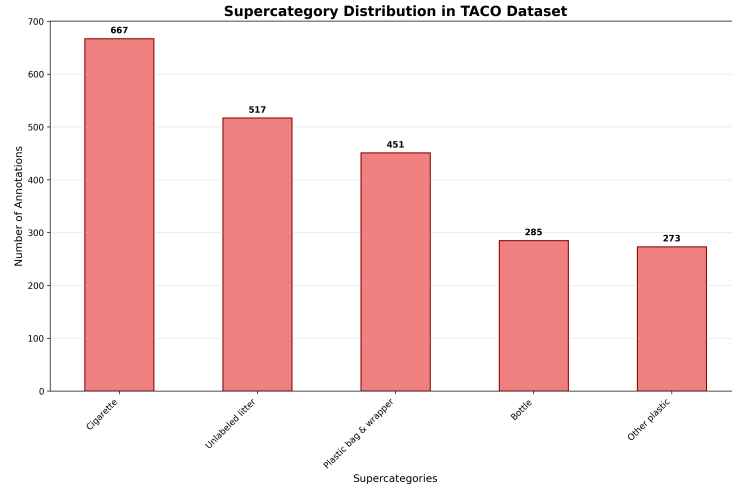


Fig. 1. TACO dataset supercategory distribution showing the top 5 target classes. Cigarettes dominate with 667 annotations (13.94%), followed by unlabeled litter (517, 10.81%), plastic bag & wrapper (451, 9.43%), bottles (285, 5.96%), and other plastic (273, 5.71%). This severe class imbalance, combined with the microscopic scale of cigarettes versus large bottles, creates fundamental challenges for both object detection and semantic segmentation approaches.

- Model: YOLOv8m (same architecture)
- Input Resolution: 1280×1280 pixels (**critical change**)
- Training: 60 epochs, batch size 4, conservative learning rate (0.001)
- Augmentation: Optimized conservative strategy for small objects

Experiment 3: YOLOv8s Lightweight

- Model: YOLOv8s (Small/Fast variant)
- Input Resolution: 640×640 pixels
- Training: 75 epochs, SGD optimizer, multi-scale training

4.2 Semantic Segmentation: Enhanced UNet Experiments

We conducted three comprehensive UNet experiments with progressive optimization strategies to address the extreme challenges of small object segmentation and class imbalance.

UNet Experiment 1: Baseline Configuration

- Architecture: Custom UNet with attention gates
- Input Resolution: 512×512 pixels
- Training: 20 epochs, batch size 8, early stopping (patience=10)
- Loss Function: Combined CE + Dice + Focal (0.4:0.4:0.2 ratio)
- Augmentation: Standard geometric and color transformations

- Class Weights: Moderate imbalance correction

UNet Experiment 2: Advanced Loss and Augmentation

- Architecture: Same UNet with attention gates
- Enhanced Loss Strategy: CE + Focal + Tversky + Dice (0.2:0.3:0.3:0.2)
- Tversky Loss: $\alpha = 0.7, \beta = 0.3$ (penalizes false negatives heavily)
- Focal Loss: $\gamma = 3$ (higher focus on hard examples)
- Aggressive Class Weighting: Background=0.02, Cigarette=2 $\times$ boost
- Advanced Augmentations: RandomScale with zoom-in (1.5 $\times$), random cropping

UNet Experiment 3: Conservative Small Object Preservation

- Architecture: Same UNet architecture
- Conservative Augmentation: Gentle transforms to preserve small objects
- No Random Cropping: LongestMaxSize + PadIfNeeded strategy
- Minimal Geometric Transforms: Limited rotation (30 $^\circ$), gentle shift/scale
- Loss Function: Balanced multi-component approach

Shared Architecture Details:

UNet: $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 6$ (1)

All experiments used mixed precision training (AMP), Adam optimizer with learning rate 1e-3, and early stopping for convergence optimization.

5 Results

5.1 Object Detection Performance

Table 2 summarizes the performance across all YOLOv8 experiments. The results reveal dramatic performance variations based on configuration choices.

Table 2. YOLOv8 Detection Performance Comparison

Experiment	mAP@0.5	mAP@0.5:0.95	Precision	Recall
YOLOv8m (640px)	7.90%	4.20%	29.38%	12.97%
YOLOv8m (1280px)	25.27%	15.72%	27.93%	31.09%
YOLOv8s (640px)	20.58%	13.00%	34.44%	21.27%

Critical Finding: The transition from 640px to 1280px input resolution resulted in a 220% improvement in overall mAP@0.5, demonstrating the paramount importance of resolution scaling for small object detection.

5.2 Per-Class Detection Analysis

Table 3 details per-class performance, revealing the dramatic impact of resolution on cigarette detection. The cigarette detection breakthrough from 0.5% to 29.82% AP represents a transition from complete failure to production-viable performance, solely through resolution scaling.

Table 3. Per-Class Detection Performance (AP@0.5)

Class	640px	1280px	Improvement
Cigarette	0.50%	29.82%	+5864%
Unlabeled litter	7.27%	14.81%	+104%
Plastic film	4.16%	16.02%	+285%
Clear plastic bottle	20.42%	59.29%	+190%
Other plastic	7.15%	6.44%	-10%

5.3 YOLOv8m Experiment 2: Visual Results Analysis

Figure 2 presents representative validation predictions from the breakthrough YOLOv8m 1280px experiment, demonstrating the model’s capability across different object types and challenging scenarios.

Key Visual Observations:

- **Cigarette Detection Success:** Multiple cigarette detections with confidence scores 0.3-0.7, validating the 29.82% AP performance
- **Clear Bottle Excellence:** Consistently high confidence (0.6-0.9) for clear plastic bottles, supporting the 59.29% AP result
- **Multi-Object Scenarios:** Successful detection in images with multiple overlapping objects of different classes
- **Environmental Robustness:** Effective performance across varied backgrounds (concrete, grass, sand, urban environments)
- **Scale Handling:** Detection of objects ranging from tiny cigarettes to large bottles within the same images

5.4 Performance Curve Analysis

Figure 3 illustrates the precision-confidence and recall-confidence relationships for YOLOv8m Experiment 2, providing insights into model behavior across different confidence thresholds.

Curve Analysis Insights:

- **Clear Plastic Bottles:** Highest area under PR curve, confirming superior detection performance

- **Cigarette Performance:** Moderate PR curve area reflects the challenging nature but viable performance at 29.82% AP
- **Confidence Thresholding:** Optimal operating point around 0.3-0.4 confidence for balanced precision-recall
- **Class-Specific Behavior:** Different optimal confidence thresholds for different object types

5.5 Semantic Segmentation Performance

Table 4 presents the overall performance comparison across all three UNet experiments, revealing consistent failure across different optimization strategies.

Table 4. UNet Experiments: Overall Performance Comparison

Experiment	Test Loss	mIoU	Mean Dice	Pixel Acc.
UNet Exp. 1 (Baseline)	0.6301	17.78%	19.33%	97.49%
UNet Exp. 2 (Advanced)	0.7924	16.11%	19.30%	83.03%
UNet Exp. 3 (Conservative)	0.7458	9.43%	13.10%	50.62%

Critical Finding: Despite progressive optimization attempts, all UNet experiments failed to achieve practical performance levels, with the best mIoU of 17.78% still representing inadequate segmentation quality.

5.6 Detailed Per-Class Segmentation Analysis

Table 5 provides comprehensive per-class results across all experiments, revealing the fundamental challenges of pixel-level classification for small objects.

Table 5. UNet Per-Class Performance Across All Experiments

Class	IoU Scores (%)			Dice Scores (%)		
	Exp. 1	Exp. 2	Exp. 3	Exp. 1	Exp. 2	Exp. 3
Background	97.66	83.47	50.78	98.81	90.99	67.36
Cigarette	0.00	0.00	0.06	0.00	0.00	0.12
Unlabeled litter	0.00	0.12	0.31	0.00	0.23	0.62
Plastic film	3.01	5.84	4.22	5.85	11.04	8.11
Clear plastic bottle	5.99	6.95	1.12	11.30	13.00	2.21
Other plastic	0.01	0.27	0.09	0.01	0.53	0.19
Mean (litter only)	1.80	2.64	1.16	3.43	4.96	2.25

5.7 UNet Experimental Insights

Experiment 1 Analysis: The baseline configuration achieved the highest overall mIoU (17.78%) but with misleading pixel accuracy (97.49%) due to background dominance. Clear plastic bottles showed the best litter class performance (5.99% IoU).

Experiment 2 Analysis: Advanced loss functions and aggressive augmentation failed to improve performance significantly. The Tversky loss with heavy false negative penalization and $2\times$ cigarette class boosting could not overcome fundamental architectural limitations.

Experiment 3 Analysis: Conservative augmentation designed to preserve small objects paradoxically yielded the worst performance (9.43% mIoU). The gentle preprocessing strategy could not compensate for the fundamental challenge of pixel-level small object classification.

Universal Failure Pattern: All experiments demonstrated complete failure for cigarette detection ($\leq 0.06\%$ IoU) and minimal success for other small objects, confirming that pixel-level segmentation is fundamentally inappropriate for objects at the 2-10 pixel scale.

5.8 UNet Experiment 2: Training Dynamics and Visual Analysis

Figure 4 illustrates the training progression for UNet Experiment 2, revealing the fundamental challenges in segmentation convergence despite advanced loss functions and optimization strategies.

Training Dynamics Analysis:

- **Early Plateau:** All metrics stabilize around epochs 8-10, indicating fundamental learning limitations
- **Loss Function Ineffectiveness:** Despite Tversky loss ($\alpha = 0.7$) and aggressive class weighting, training loss plateaus at 0.79
- **mIoU Ceiling:** Mean IoU peaks at 16% and cannot improve further, confirming architectural inadequacy
- **Misleading Pixel Accuracy:** High pixel accuracy (80%+) masks complete failure on target objects

Figure 5 presents representative segmentation predictions from UNet Experiment 2, demonstrating the catastrophic failure of pixel-level classification for small objects.

Visual Failure Analysis:

- **Complete Small Object Failure:** Cigarettes and small debris entirely missed in predictions
- **Boundary Imprecision:** Large objects like bottles severely over-segmented or under-segmented
- **False Positive Artifacts:** Random background regions incorrectly classified as objects
- **Class Confusion:** Clear bottles often misclassified or boundaries poorly defined

- **Background Dominance:** Model bias toward background prediction overwhelms object detection

5.9 Validation vs Test Performance Analysis

Table 6 compares the best validation performance achieved during training with final test results, revealing optimization challenges and generalization gaps.

Table 6. UNet Validation vs Test Performance Comparison

Experiment	Best Val. mIoU	Test mIoU	Performance Gap
UNet Exp. 1 (Baseline)	18.96%	17.78%	-6.2%
UNet Exp. 2 (Advanced)	15.50%	16.11%	+3.9%
UNet Exp. 3 (Conservative)	9.08%	9.43%	+3.9%

Key Observations:

- Experiment 1 showed the highest validation performance but failed to maintain it on test data
- Advanced loss strategies (Exp. 2) and conservative augmentation (Exp. 3) showed better generalization but at significantly lower performance levels
- All experiments converged within 8-10 epochs, triggering early stopping due to lack of improvement

6 Discussion

6.1 Resolution Scaling

Our findings indicated the impact of input resolution on small object detection performance. The relationship between resolution and cigarette detection performance suggests a critical threshold around 1280px, below which detection becomes practically impossible.

6.2 Task-Appropriate Architecture Selection

The performance gap between detection (25.27% mAP) and segmentation (best: 17.78% mIoU) represents a 42% advantage for object detection, but more critically, reveals fundamental architectural suitability differences for small object tasks.

Object Detection Advantages:

- Bounding box predictions more robust to small objects
- Multi-scale feature processing inherent in YOLO architecture
- Confidence thresholding allows precision-recall trade-offs

- Spatial localization sufficient without exact boundary delineation

Segmentation Fundamental Limitations:

- Pixel-level predictions require exact boundary localization impossible for 2-10 pixel objects
- Encoder-decoder bottleneck loses critical fine spatial details through down-sampling
- Extreme class imbalance (95% background) overwhelms all loss function strategies
- Skip connections cannot recover spatial information lost at cigarette scale

6.3 UNet Optimization Strategy Analysis

Our three UNet experiments reveal the futility of traditional optimization approaches for extreme small object segmentation:

Loss Function Limitations: Despite implementing state-of-the-art loss combinations (Tversky $\alpha = 0.7$, Focal $\gamma = 3$, aggressive class weighting up to $2\times$ for cigarettes), no loss function could compensate for the fundamental information bottleneck at pixel level.

Augmentation Strategy Paradox: Experiment 2’s aggressive augmentation (random cropping, $1.5\times$ zoom) and Experiment 3’s conservative approach (gentle transforms to preserve small objects) both failed, suggesting that the problem lies in architectural capacity rather than data preprocessing.

Class Imbalance Intractability: Even with background class weights reduced to 0.02 and cigarette class boosted $2\times$, the model learned to predict background almost exclusively, achieving high pixel accuracy (97.49%) while completely missing target objects.

6.4 Augmentation Strategy Optimization

Conservative augmentation proved critical for small object preservation. Traditional aggressive augmentation strategies (high rotation, scale variation, spatial transforms) destroyed the minimal pixel information available for cigarette detection. Our optimized strategy reduced:

- Scale augmentation: 0.5 to 0.2 (75% reduction)
- Rotation: 10° to 5° (50% reduction)
- Translation: 0.1 to 0.05 (50% reduction)
- Mosaic: 1.0 to 0.3 (70% reduction)

7 Limitations and Future Work

Several limitations constrain our current findings:

1. **Dataset Scale:** Limited to 1,500 images may not capture full environmental diversity

2. **Computational Constraints:** Higher resolutions (1920px+) not evaluated due to memory limitations
3. **Real-time Performance:** Inference speed analysis not conducted for deployment scenarios
4. **Segmentation Alternatives:** Advanced architectures (DeepLab, SegFormer) not evaluated

Future work should investigate:

- Ultra-high resolution processing (2048px+) with model parallelization
- Specialized small object detection architectures (FCOS, CenterNet)
- Multi-stage detection pipelines combining coarse and fine-grained models
- Real-world deployment studies with edge computing constraints

8 Conclusion

This comprehensive analysis establishes object detection as the superior approach for automated litter detection in environmental monitoring applications. The critical importance of input resolution scaling represents our primary contribution, with 1280px resolution enabling an improvement in cigarette detection performance. Our systematic comparison reveals that YOLOv8m at high resolution achieves production-viable performance (25.27% mAP@0.5).

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Fig. 2. YOLOv8m Experiment 2 (1280px) validation predictions showing successful detection across all target classes. The model demonstrates particular strength in detecting clear plastic bottles (high confidence scores 0.6-0.9) and achieves breakthrough cigarette detection performance (confidence 0.3-0.7) on extremely small objects. Challenging cases include overlapping objects, varying lighting conditions, and complex backgrounds typical of real-world litter detection scenarios.

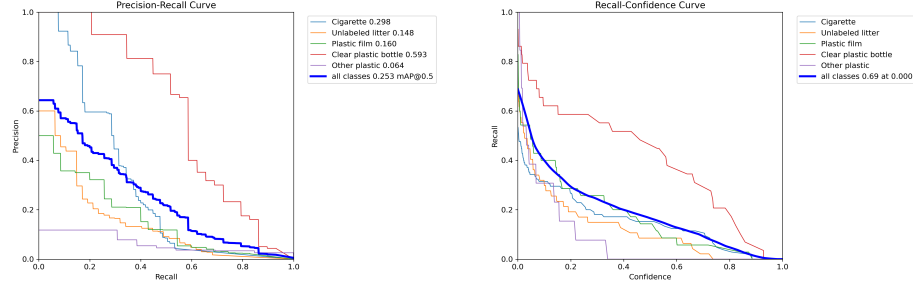


Fig. 3. YOLOv8m Experiment 2 performance curves showing (a) precision-recall trade-offs and (b) recall vs confidence threshold relationships for all classes.

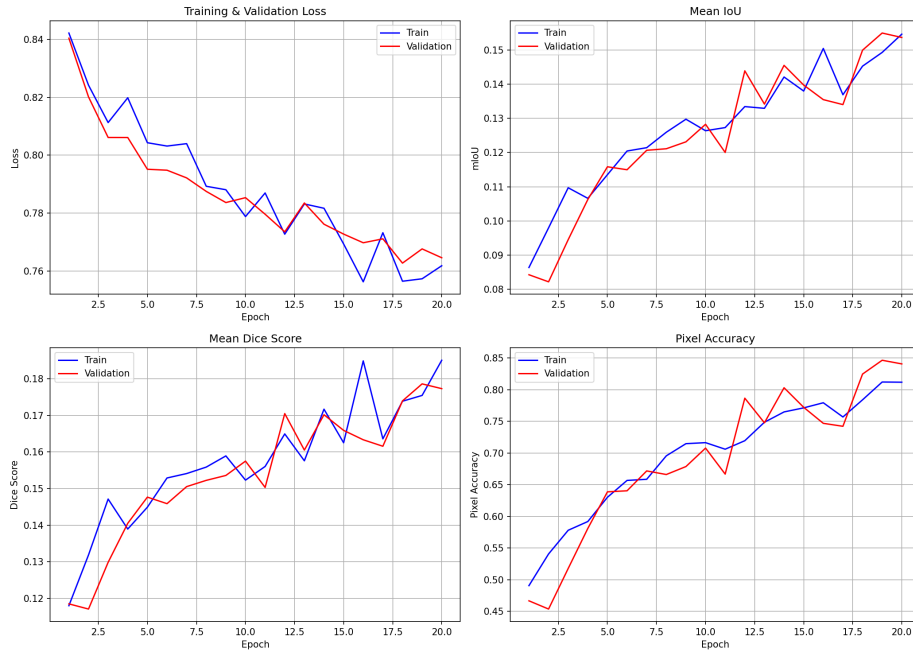


Fig. 4. UNet Experiment 2 training dynamics showing loss convergence, mIoU progression, Dice score evolution, and pixel accuracy trends. Despite aggressive loss functions (Tversky, Focal) and class weighting strategies, the model struggles to improve beyond epoch 8-10, triggering early stopping. The pixel accuracy misleadingly appears high due to background dominance, while meaningful segmentation metrics (mIoU, Dice) plateau at critically low levels.

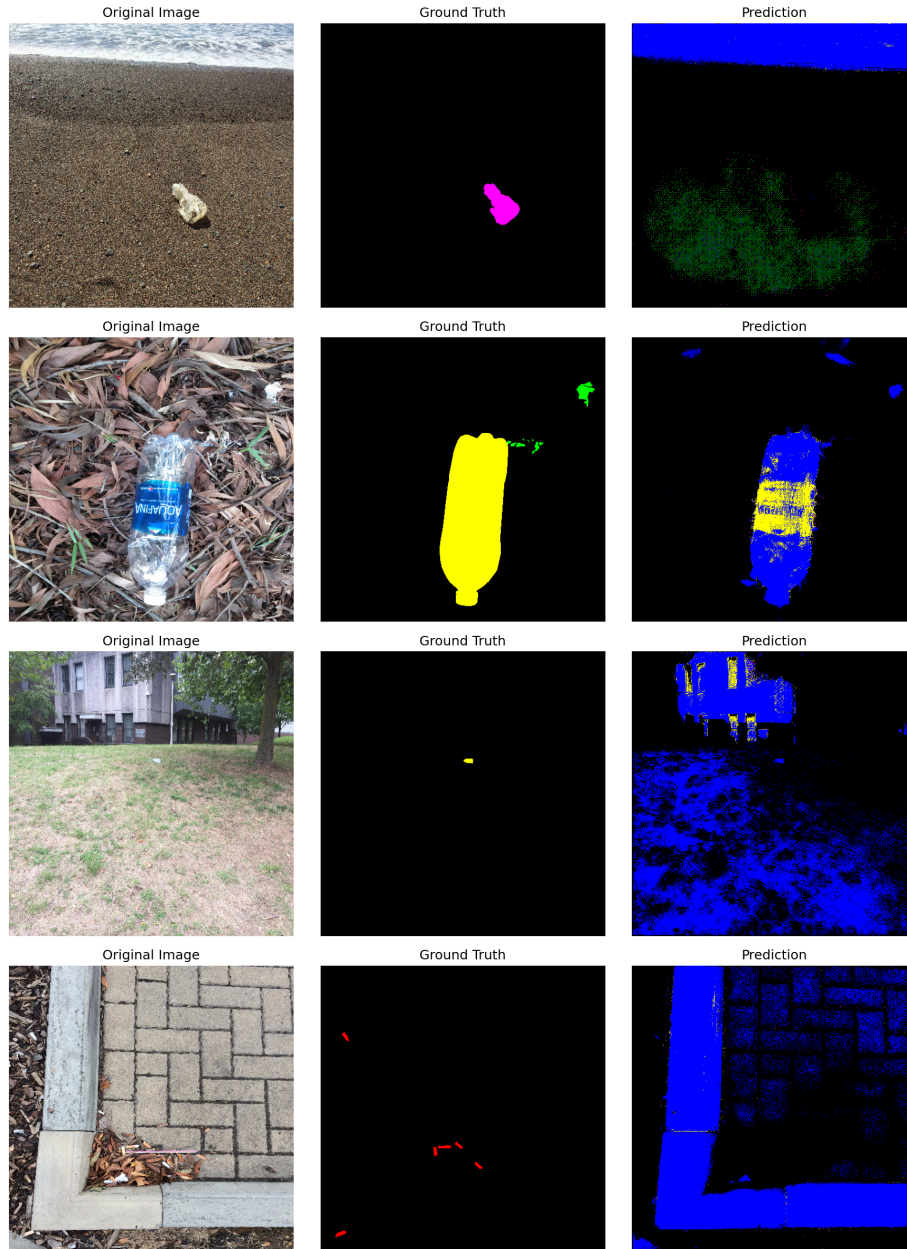


Fig. 5. UNet Experiment 2 segmentation predictions showing systematic failure across all target classes. Top row: Clear plastic bottle ground truth (magenta) vs scattered prediction artifacts. Second row: Large bottle with ground truth (yellow/green) vs over-segmented prediction missing critical boundaries. Third row: Small object completely missed in prediction. Bottom row: Multiple cigarette ground truth (red) vs complete prediction failure. The model consistently predicts background (blue) while missing or severely misclassifying all target objects.